Information Processing Letters 110 (2010) 113–115

Contents lists available at ScienceDirect

Information Processing Letters

www.elsevier.com/locate/ipl

Finding the maximum bounded intersection of k out of n halfplanes

Peter Brass [a], Hyeon-Suk Na [b,*,1]

[a] Department of Computer Science, City College, New York, NY, USA
[b] School of Computing, Soongsil University, Seoul, Republic of Korea

ARTICLE INFO

Article history:
Received 25 September 2008
Received in revised form 27 October 2009
Accepted 2 November 2009
Available online 13 November 2009
Communicated by F.Y.L. Chin

Keywords:
Computational geometry
Algorithms
Maximum intersection

ABSTRACT

In this note we study the following question: Given n halfplanes, find the maximum-area bounded intersection of k halfplanes out of them. We solve this problem in $O(n^3k)$ time and $O(n^2k)$ space.

1. Introduction

The question for the largest-area polygon spanned by k out of n points is well-studied, it has been solved in $O(n(k + \log n))$ time [2], improving [4,5], and for sorted points even in $O(n2^{O(\sqrt{\log k \log\log n})})$ for $k = \Omega(\log n)$ [3,8]. The question for the largest-area polygon formed by the intersection of k out of n halfplanes looks in some way dual, but turns out to be much harder. For the points in the plane, convexity generates a cyclic order, but a similar structure is missing for the lines or halfplanes. To our best knowledge, the only relevant problem that has been studied before is the smallest circumscribing k-gon problem for n lines, all touching some convex set and thus spanned by the sides of a convex n-gon. This problem has been studied by similar methods in [1,2,6,7] and ultimately has arrived at the same bounds [3,8]. But without this convex position assumption of the lines, the problem has not been studied before.

In the problem statement, we need to be somewhat careful, since among any system of n halfplanes there are at least $\frac{1}{2}n$ halfplanes whose outer normals lie in the same π-sector of direction space and thus whose intersection is unbounded. We are looking for the largest-area *bounded* intersection of k out of n halfplanes (see Fig. 1 for an example). Here "largest" is interpreted as largest area, but the algorithm easily adapts to largest perimeter, or any size measure with a suitable piecewise algebraic parametrization. The bounded intersection of k halfplanes is not necessarily a k-gon but might have less sides; again an easy modification adapts to find the largest k-gon.

Theorem 1. *Given n halfplanes, we can find the maximum-area bounded intersection of k halfplanes out of them, in $O(n^3k)$ time and $O(n^2k)$ space.*

2. The algorithm

Given n halfplanes $H_1, \ldots, H_n$ with their bounding lines $\ell_1, \ldots, \ell_n$, we want to find k distinct halfplanes $H_{i_1}, \ldots, H_{i_k}$ such that $H_{i_1} \cap \cdots \cap H_{i_k}$ is bounded and has maximum-area among all bounded intersections of k halfplanes.

Let $H(t)$ be the halfplane of points with x-coordinate not greater than t, and let $\ell(t)$ be its bounding line. De-

* Corresponding author.
E-mail addresses: peter@cs.ccny.cuny.edu (P. Brass), hsnaa@ssu.ac.kr (H.-S. Na).

1 Supported by Korean Research Foundation Grant (KRF-2007-531-D00018).

0020-0190/$ – see front matter
doi:10.1016/j.ipl.2009.11.003

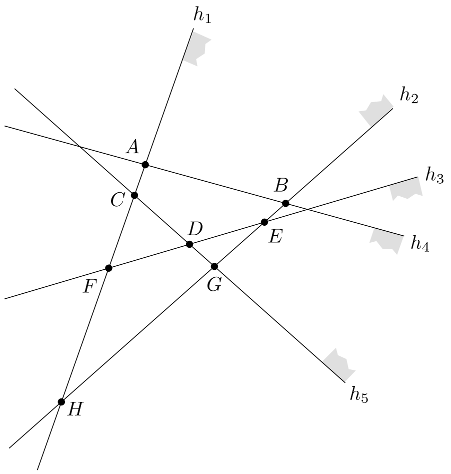

Fig. 1. Finding the largest-area polygon formed by the intersection of k out of five halfplanes $h_1, \ldots, h_5$. The optimal polygon for $k = 3$ is $\triangle ABH = h_1 \cap h_2 \cap h_4$, and the optimal for $k = 4$ is $\triangle FEH = h_1 \cap h_2 \cap h_3 \cap h_4$; any other intersection of four halfplanes including $h_1 \cap h_2 \cap h_4 \cap h_5$ is either unbounded or smaller than this triangle. For $k = 5$, the only choice is the intersection of all halfplanes, $\triangle DEG$.

fine $A_t(a, b, c)$ as the maximum-area of any bounded intersection of $H(t)$ with c out of $H_1, \ldots, H_n$, among which H_a, $H(t)$, and H_b occur consecutively in the clockwise ordering of $(c+1)$ outer normals. So $A_t(a, b, c)$ is the area of some polygon bounded by $\ell_a, \ell(t), \ell_b$, and $c-2$ other line-segments to the left of $\ell(t)$, or 0 if no such polygon exists. This area depends on t, since we cut off some polygon or open cone by the line $\ell(t)$; it is a piecewise quadratic function of t whose coefficients only change when $\ell(t)$ passes an intersection point $\ell_i \cap \ell_j$ with $\{i, j\} \cap \{a, b\} \neq \emptyset$.

We now describe our algorithm. As a preprocessing step, we generate all intersection points $\ell_i \cap \ell_j$ and sort them by x-coordinate in nondecreasing order, and for each intersection point, we compute how many halfplanes H_r satisfy $H_i \cap H_j \subseteq H_r$. This can be done in $O(n^3)$ time and requires $\Theta(n^2)$ space.

We then perform a sweep with $\ell(t)$ over the line-arrangement and maintain $A_t(a, b, c)$ (by its coefficients) for all (a, b, c) that $1 \leqslant a, b \leqslant n$ and $2 \leqslant c \leqslant k$. Whenever the sweep line $\ell(t)$ passes an intersection point $\ell_i \cap \ell_j$, we have to update the coefficients of $A_t(a, b, c)$ for some triples (a, b, c) that $\{i, j\} \cap \{a, b\} \neq \emptyset$ and $2 \leqslant c \leqslant k$. For instance, if we pass at moment t_0 the intersection point $\ell_i \cap \ell_j$ with $H_i \cap H_j \subseteq H(t_0)$, as in Fig. 2(a), and this intersection point has pre-computed information that there are d other halfplanes containing $H_i \cap H_j$, then the numbers $A_{t_0}(i, j, c-d), \ldots, A_{t_0}(i, j, c)$, if they are nonzero, represent areas of bounded intersection of c halfplanes and the

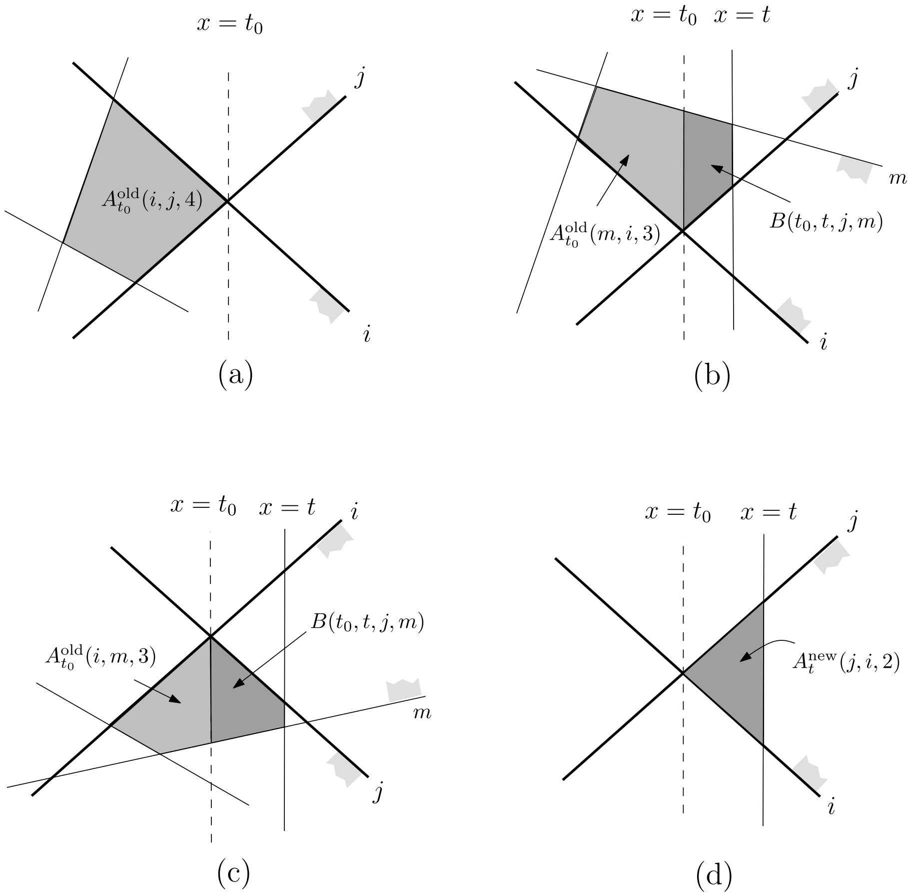

Fig. 2. Illustration of the four cases when the sweep line passes the point $\ell_i \cap \ell_j$.

function $A_t(i,j,c)$ must be updated by their maximum. Also, the new $A_t(i,j,k)$ is a candidate for the maximum-area bounded intersection of k halfplanes and must be compared with the previous best candidate.

Now we give the update rules for $A_t(a,b,c)$ in more details: Initially all these functions are set to be constant 0. Let $B(s,t,i,j)$ be the area of the trapezoid (set to be 0 if unbounded) defined by the intersection $H_i \cap H_j$ and the vertical lines $x=s$ and $x=t$. Suppose that at moment t_0 the line $\ell(t_0)$ passes the intersection $\ell_i \cap \ell_j$ and there are d other halfplanes containing $H_i \cap H_j$. The wedge $H_i \cap H_j$ can lie in four possible situations relative to $\ell(t_0)$.

(a) $H_i \cap H_j$ lies entirely to the left of $\ell(t_0)$. In that case all polygons starting with H_i and ending with H_j in the counter-clockwise order are closed and do not change their area anymore. Thus we have

$$A_t^{\text{new}}(i,j,c) = \max_{r \leqslant d} A_{t_0}^{\text{old}}(i,j,c-r).$$

Here $A_t^{\text{new}}(i,j,k)$ is a candidate for the maximum-area bounded intersection of k halfplanes and must be compared with the previous best candidate. No other functions $A_t(a,b,c)$ are updated.

(b) $H_i \cap H_j$ is intersected in the interior by $\ell(t_0)$ and extends upward. We can assume that $\ell_i \cap H_j$ lies left of $\ell(t_0)$ and $\ell_j \cap H_i$ lies right of $\ell(t_0)$. Then only those functions of the form $A_t(m,j,c)$ change for $t \geqslant t_0$, where ℓ_m intersects $\ell(t)$ above ℓ_j with H_m extending downward. For them, we update

$$A_t^{\text{new}}(m,j,c) = \max\Big(A_t^{\text{old}}(m,j,c), \max_{r \leqslant d} A_{t_0}^{\text{old}}(m,i,c-1-r) + B(t_0,t,j,m)\Big).$$

(c) $H_i \cap H_j$ is intersected in the interior by $\ell(t_0)$ and extends downward. We assume that $l_i \cap h_j$ lies left of $\ell(t_0)$ and $l_j \cap h_i$ lies right of $\ell(t_0)$. Then only those functions of the form $A_t(j,m,c)$ change for $t \geqslant t_0$, where ℓ_m intersects $\ell(t)$ below ℓ_j with H_m extending upward. For them, we have

$$A_t^{\text{new}}(j,m,c) = \max\Big(A_t^{\text{old}}(j,m,c), \max_{r \leqslant d} A_{t_0}^{\text{old}}(i,m,c-1-r) + B(t_0,t,j,m)\Big).$$

(d) $H_i \cap H_j$ lies entirely to the right of $\ell(t_0)$. Then

$$A_t^{\text{new}}(j,i,c) = B(t_0,t,i,j) \quad \text{for } c \leqslant \min(d+2,k).$$

No other functions $A_t(a,b,c)$ are updated.

At the end of the sweep, the largest candidate for the maximum gives the maximum-area bounded intersection of k halfplanes out of n.

There are $\Theta(n^2)$ intersection points (events) in total, and at each intersection point $\ell_i \cap \ell_j$ we update the coefficients of $O(nk)$ functions $A_t(a,b,c)$ such that $\{i,j\} \cap \{a,b\} \neq \emptyset$. Therefore the sweeping step requires in total $O(n^3k)$ time and $O(n^2k)$ space, and so does the whole algorithm.

3. Further comments

The maximum bounded intersection problem seems unexpectedly difficult; we have neither a fast approximation, nor any extension to higher-dimensional analogues. But the problem of the largest spanned k-gon out of n points also has not been solved in higher dimensions. An interesting abstract version of our problem is obtained if we have one point p inside the maximum-area intersection, and follow a ray that rotates around p; this ray intersects each halfplane boundary at some distance, or not at all. Thus we obtain n partial functions $f_1,\ldots,f_n$ defined on $[0,2\pi)$, and we wish to select k functions $f_{i_1},\ldots,f_{i_k}$ such that $\min\{f_{i_1}(\varphi),\ldots,f_{i_k}(\varphi)\}$ is defined for each $\varphi \in [0,2\pi)$, and among all such k-tuples the integral $\int_0^{2\pi} \min\{f_{i_1}(\varphi),\ldots,f_{i_k}(\varphi)\}\,d\varphi$ is maximal. The analogue to our dynamic-programming algorithm in this setting requires that any two functions intersect at most once.

References

[1] A. Aggarwal, J.S. Chang, C.K. Yap, Minimum area circumscribing polygons, Visual Computer 1 (1985) 112–117.

[2] A. Aggarwal, M. Klawe, S. Moran, P. Shor, R. Wilber, Geometric applications to a matrix searching algorithm, Algorithmica 2 (1987) 209–233.

[3] A. Aggarwal, B. Schieber, T. Tokuyama, Finding a minimum-weight k-link path in graphs with the Monge property and applications, Discrete & Computational Geometry 29 (1998) 204–222.

[4] J.E. Boyce, D.P. Dobkin, R.L. Drysdale, L.J. Guibas, Finding extremal polygons, SIAM Journal of Computing 14 (1985) 134–147.

[5] D.P. Dobkin, L. Snyder, On a general method for maximizing and minimizing among certain geometric problems, in: FOCS 1979, 20th IEEE Symposium on Foundations of Computer Science, pp. 9–17.

[6] V. Klee, M. Laskowski, Finding the smallest triangles containing a given convex polygon, Journal of Algorithms 6 (1985) 359–375.

[7] J. O'Rourke, A. Aggarwal, S. Maddila, M. Baldwin, An optimal algorithm for finding minimum enclosing triangles, Journal of Algorithms 7 (1986) 258–269.

[8] B. Schieber, Computing a minimum-weight k-link path in graphs with the concave Monge property, Journal of Algorithms 29 (1998) 204–222.